

BERTopic 활용 금융앱 UI/UX 요소와 슈퍼앱 출시 전후 사용자 반응 변화분석: 국내 대표 금융그룹 중심으로*

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Analyzing UI/UX Elements of Financial Apps Using BERTopic and Changes in User Reactions before and after the Super App Launch: Focusing on a Leading Korean Financial Group*

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■ Abstract ■

This study investigates how user experience (UX) and user Interface (UI) is influenced by the adoption of financial super apps. As major Korean financial groups have launched integrated financial super apps that consolidate various services, they are now moving toward further enhancing and sophisticating these platforms. The study aims to empirically analyze user reactions before and after the launch of a super app. User reviews were collected from the Google Play Store, covering financial superapp, pay app, trading app, insurance app, and banking app as non-superapp. To extract meaningful insights from unstructured textual data, this study applied BERT-based topic modeling technique—combined with ordinary least squares (OLS) regression to quantify the influence of topic on user ratings. Ten core UI/UX-related topics were identified: identity verification service, stock trading function, simple payment service, event, menu UI, ID card recognition function, home screen UI, daily participatory interaction, app icon design, and the sign-up process function. All topics had statistically significant effects on satisfaction; however, the results indicate that the super app environment tended to heighten user expectations, which in turn contributed to reduced satisfaction ratings across topics. Additionally, by analyzing changes in topic proportions before and after the launch of the super app, the topics were classified into three distinct groups based on their temporal dynamics. Group 1 includes features that remained influential even in non-super app environments. Group 2 represents topics that became more prominent in the super app context, reflecting the added value brought by integration and expanded interaction. Group 3 consists of topics that consistently influenced user satisfaction across both super app and non-super app environments, regardless of the platform transition. This study contributes to a deeper understanding of UI/UX in financial super apps and provides practical insights.

Keyword : Super-App, Online Review, Topic Modeling, Satisfaction, OLS, BERTopic

Submitted : June 16, 2025

1st Revision : August 18, 2025

Accepted : August 20, 2025

* This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF-2018S1A5A2A01039413)

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1. Introduction

In recent times, domestic financial groups have initiated the development of ‘super-apps’, which are characterised by the integration of diverse functionalities into a single interface. Super apps achieve their ‘super’ status by integrating a wide range of essential everyday services across industries and borders within constellation types (Vlist et al., 2024). One of the best examples of a superapp is ‘GOJEK’. ‘GOJEK’ started as a motorcycle taxi booking service in Indonesia, but has since expanded to incorporate a variety of services including transportation, logistics, food delivery, payments, and more (Jin et al., 2023). This strategy has made it a successful superapp in Indonesia and across Southeast Asia (Ko, 2023).

Leading financial groups in Korea are accelerating their digital financial transformation with superapp strategies to ensure user-centric convenience and integration. In the past, each affiliate operated its own app, but with the rise of the “one app” approach that offers a variety of services on a single platform, there is a clear trend toward building super apps to gain a competitive advantage (Jeong, 2023). In addition, in response to the rapidly changing digital environment, the company is increasing its competitiveness by adding services from affiliated companies to one app and equipping it with various life services, and aims to build a comprehensive financial life platform by combining financial and non-financial services (Han, 2022). In addition, the aim is to increase customer usage by offering a comprehensive range of services and meeting customer needs through the services provided by subsidiaries under the holding company (Ji, 2022).

Looking at the adoption trend of super apps

by major Korean traditional financial institutions, H Financial Group was the first to launch a super app in August 2020, followed by K Financial Group in October 2021 with an initial model of the super app and then an advanced version in October 2022. S Financial Group followed in December 2023, W Financial Group in December 2024, and finally N Financial Group in February 2025, completing the launch of all major Korean traditional banks’ super apps. This trend is expected to intensify not only competition from non-traditional financial companies, but also competition among traditional financial companies.

As of June 5, 2025, the user ratings of financial super apps of major traditional financial groups in Korea are as follows <Table 1>.

<Table 1> Ratings of Financial Superapps from Major Traditional Financial Institutions in Korea

Institution	Google Play Store	App Store
S	4.6	4.4
K	4.6	4.3
H	4.3	4.3
W	3.4	4.2
N	3.7	2.4

* Google Play Store, AppStore Review(‘25.6.5).

S Financial Group’s super app, which has been ranked as the No. 1 bank in Korea for 15 consecutive years in the Korea Service Quality Index (KSQI), also has the highest satisfaction among consumers, according to user star ratings on the Google Play Store and App Store. Although these star ratings show that S’s super apps are receiving the most positive responses from users, it is difficult to identify specific factors that positively affect user satisfaction based on simple star ratings. In addition, the super app of S Financial Group

is more meaningful to analyze because it is not like K Financial Group and H Financial Group, which are ranked second and third, which developed the existing bank's app as a super app and provided all the functions of the existing bank's app and the essential functions of other affiliates, but only the essential functions of all affiliate functions are collected and provided as a new app. Therefore, it was necessary to analyze actual user reviews to identify specific UI/UX factors that affect user satisfaction. Based on user review data, this study derived UI/UX-related topic elements that affect user satisfaction in S Financial Group's apps that are positively evaluated, and identified the influence of each element through statistical analysis.

In addition, by analyzing the change in the proportion of reviews by topic before and after the introduction of the super app, we wanted to understand the impact of the super app launch on the actual user experience. In this research, user reaction is defined as an explicit action performed by the user through a widget to express her/his actual or expected interest, evaluation, or emotion with respect to system's objects (Cena et al., 2021). These reactions are expressed in the form of star ratings, emojis, text reviews, and more, and are key clues to indirectly infer a user's satisfaction or experience. To do this, we first collected text reviews and star ratings of S Financial Group's super apps and non-super apps (securities app, insurance app, banking app, and pay app) on Google Play Store. After preprocessing the collected data, we extracted 10 UI/UX-related topics using the BERTopic (Bidirectional Encoder Representations from Transformers Topic) model, a topic modeling technique. We then regressed the correlation between topics and star ratings

to determine their impact on satisfaction, and analyzed the temporal trends in the proportion of reviews by topic to examine the changes in user experience before and after the release of the super app. We were able to categorize each topic into three types based on how the weight of each topic changed, and analyzed how the topics in each group affected users. Through this analysis, we aimed to empirically identify which topics were the factors that caused users to be satisfied or dissatisfied, and how the switch to super apps changed the user experience.

2. Theoretical background

2.1 Research on Super Apps

The term superapp was popularized by BlackBerry founder Mike Lazaridis, who defined a superapp as a closed ecosystem offering a seamless, integrated, and efficient user experience. Global research firm Gartner (2022) describes superapps as an application that provides end users with a set of core features and access to independently created miniapps. Research firm Deloitte (2022) defines a superapp as a single mobile application that integrates a variety of services typically offered by multiple, separate apps.

Super apps emerged to provide multiple services in one app as the number of apps needed to be installed to use various services increased and the inconvenience of having to sign up repeatedly to use the app increased. These super apps can be viewed as portals to the app space, and the multiple mini apps embedded in the super app are the key to making it possible to use various services without installing separate apps (Her, 2023).

The definition and structural characteristics of superapps are also being debated in international research. Lucas Júnior and Lopes (2024) conducted a Systematic Literature Review of 26 relevant articles published between 2018 and 2023 and redefined a superapp as “an open platform for partners to develop and deploy mini-apps.” According to the study, a superapp is not just a tool that integrates various functions into one app, but a digital ecosystem with three core elements: (1) mini-app integration structure, (2) operation based on strategic partnerships, and (3) scalability by utilizing user data. In particular, we emphasize that partner linkages and data-driven service expansion promote network effects, and that super apps are evolving beyond a single service provider into a multi-layered platform ecosystem. This definition and analysis provides an important conceptual foundation for the study of superapp expansion strategies and platform competitiveness. Furthermore, Hasselwander and Weiss (2025) analyzed the preferences of European (mainly German) consumers for superapp services and reported that e-commerce, social media, and financial (banking) features were particularly preferred. These results provide empirical support for why quantitative comparisons of UX responses across features are important in this study.

Noting that these super-apps integrate a wide range of features and can collect extensive behavioral data on users, there are studies that have analyzed their potential to complement traditional financial institutions’ credit scoring methods. Roa et al. (2021) applied graphical modeling of user interactions (P2P remittances, card sharing, location information, etc.) to credit risk prediction based on data from Latin American super apps.

In particular, they built a model based on XGBoost and Graph Neural Network (GNN), using user network centrality, average behavior of neighboring users, and geo-based information as features. The results showed that it is possible to achieve meaningful statistical performance and financial profitability in credit risk prediction without traditional financial data. This study empirically demonstrates that super apps can contribute to creditworthiness by utilizing social interactions and ecosystem-based data beyond simple transaction records, and also points to the potential for expanding financial inclusion. However, the study focuses on quantitative performance analysis after constructing user interaction graphs based on internal data of the super app, and lacks analysis from the user experience perspective through actual user reviews. In addition, UX satisfaction or dissatisfaction factors by function, and changes in user responses due to changes in app structure are not considered.

Şimşekler (2024) also exploratively analyzed the impact of service design elements of super apps on brand perception and user experience, and identified how core service elements such as UI, payment, customer support, returns, and logistics affect user responses. The study categorized user experience into concepts such as “Trustworthiness,” “Seamlessness,” “Functionality,” “Variability,” and “Habituality,” and explained how each element contributes to brand perception and continued use. This study provides important implications as it empirically demonstrates that the service design of a super app can have a real impact on the emotional connection and trust between users and brands. Nevertheless, while the study’s qualitative analysis-oriented research design is advantageous for deeply contextualizing

the user experience, it is limited in its ability to quantitatively compare how responses vary across features or capture structural changes over time. In addition, the data used in the study is based on researcher-collected interviews, rather than spontaneously generated reviews in real-world use, which may not reflect the frustrations or acceptance factors that occur in real-world use.

As such, super apps are not just a tool for integrating services, but have the potential to expand in various financial and non-financial areas, such as credit scoring and brand recognition based on user data. In light of these characteristics, super apps have been actively adopted in the financial sector in recent years, leading to a movement to integrate various financial services on one platform. Deloitte (2024) defines a financial super-app as an app that offers a wide range of financial services and products within a single platform, including banking, payments, investments, insurance, lending and wealth management. Also it aims to provide a seamless user experience by integrating various financial services, reducing the need for customers to use multiple apps for their financial needs.

Research on financial superapps has only recently begun to take off, and it goes something

like this. Using review data from a financial superapp, Nam et al. (2025) performed sentiment analysis, time series outlier detection, and LDA topic modeling to identify satisfaction trends and specific causes of satisfaction changes. Yoo and Song (2024) analyzed Pay Super App, which combines various elements beyond basic Pay App functions, and analyzed user reviews before and after the conversion to Pay Super App in October 2021. They organized the associations between topics derived through LDA-based topic modeling into a network structure, and derived strategic insights through a two-dimensional matrix (importance-utility) that combines the centrality index of each topic (eigenvector centrality) and the number of review recommendations (usefulness).

As a result, systematic analysis of unstructured textual data such as financial superapp user reviews requires analytical techniques that enable topic extraction and semantic structuring. In this study, we use topic modeling as a method to identify key user experience factors and analyze their evolution.

2.2 Topic Modeling

Topic modeling is the grouping of words by determining how similar they are to each other

〈Table 2〉 Research on Super Apps

Researchers	Research subjects	Analytics resources	Analytics techniques
Roa et al.	Latin American Super App users	P2P, Card, Geolocation	GNN, XGBoost
Simsekler	Global Super App	User Interviews	Qualitative analysis
Lucas et al.	Global Super App	Academic Papers	Systematic Literature Review
Hasselwander et al.	Europe Super App	Surveys	Clustering, Logistic Regression
Yunjin Nam et al.	Korea Financial Super App	Time Series Reviews	LDA, Sentiment Analysis
Jewon Yoo et al.	Pay Super App	User Reviews	LDA, Network analysis
This research	Korea Financial Super App	User Reviews	BERTopic, Logistic Regression

based on how often they co-occur in a document (Kim et al., 2023; Pak et al., 2021). It also refers to techniques that can identify a small number of meaningful and significant units, or topics, within a large set of textual documents and analyze how they are related to each other and how they change over time (Lee, 2023). In other words, topic modeling is a methodology that attempts to find the distribution of topics in a given set of documents, assuming that documents are a collection of different topics and each topic is a probabilistic distribution of words (Park, 2016). Techniques commonly used in topic modeling research include Principal Component Analysis (PCA), Latent Semantic Analysis (LSA), and Latent Dirichlet Allocation (LDA). Among the most widely used methods are Latent Semantic Analysis (LSA), which applies singular value decomposition to the document-term matrix to reduce dimensionality and capture the underlying semantic structure of the data, and Latent Dirichlet Allocation (LDA), a probabilistic topic model that represents each document as a mixture of topics, with each topic characterized by a distribution over words (Bergamaschi and Po, 2015).

However, traditional methodologies have some limitations (Choi, 2024). Firstly, topic modeling methods based on unsupervised machine learning often suffer from data sparsity when

applied to documents with a small number of words. This lack of sufficient word co-occurrence makes it difficult for traditional models to extract semantically coherent topics, resulting in the generation of ambiguous topics (Akash and Chang, 2024). Secondly, in LDA-based topic modeling, LDA incorporates a strong independence assumption rooted in the Dirichlet distribution. This has been criticized for failing to adequately capture the inter-topic correlations frequently observed in real-world documents (Blei and Lafferty, 2006). Thirdly, different settings of hyper-parameters when performing LDA can lead to different results of topic modeling, which hinders the consistency of the study. In addition, there have been criticisms that arbitrariness is involved in the process of morphological analysis, word identification, and topic derivation for optimal results (Kim and Kim, 2022).

To overcome these limitations, this study uses the BERTopic model. The BERTopic model is a topic modeling that works based on the BERT technique announced by Google in 2018, and the BERT technique is evaluated to have excellent performance in the field of natural language processing. The BERTopic model also utilizes pre-trained word embeddings and class-based TF-IDF (Class-based Term Frequency – Inverse Document Frequency) to generate dense clusters

〈Table 3〉 BERTopic Usage Study Comparison

Researchers	Research subjects	Analytics resources	Analytics techniques
Chaeyoung Hong et al.	Nuclear awareness	News Comments, Video subtitle	BERTopic, Sentiment Analysis
Boram Kwon	Generative AI Research Trends	Academic keyword	BERTopic, Keyword Network
Daun Choi et al.	Naver / Kakao Webtoon	User Reviews	BERTopic, TF-IDF
This research	Korea Financial Super App	User Reviews	BERTopic, Logistic Regression

for fast and diverse topic interpretation (Kim se-hyoung et al., 2023). In particular, word embedding techniques have attracted a lot of attention in natural language processing research due to their high accuracy and fast computational speed. Word embeddings represent words as low-dimensional vectors, and the distance between word vectors can be used to capture the semantic properties of words (Lai et al., 2015).

Given the semantic inconsistency, difficulty in reflecting inter-topic relationships, and arbitrary interventions noted in existing methodologies, the BERTopic model, which more faithfully reflects contextual information, can be seen as a suitable alternative for analyzing short and complex document sets such as financial services reviews. On the other hand, BERTopic models have recently been applied to research in various fields. For example, Hong et al. (2024) applied BERTopic-based topic modeling to analyze the differences between experts' and the public's perceptions of nuclear energy, focusing on comparing the differences in perception structure between the two groups. In addition, Kwon (2025) used keyword networks and BERTopic to structure the time-series flow of research topics and changes in central topics based on academic data related to domestic-generated artificial intelligence.

These studies are significant in that they show the applicability and scalability of the BERTopic technique and are useful analytical tools for understanding the structure of opinions and research flow on specific topics. However, there is a lack of research on user review analysis using BERTopic in the context of financial services, especially in the context of highly integrated platforms such as super-apps, as existing studies have mainly focused on identifying discourse

structure within a static analysis framework or structuring academic information.

2.3 Topic Probabilistic Regression Analysis

This feature of BERTopic is extensible in that it can not only derive topics, but also numerically utilize the topic probabilities assigned to each document. The topic probability derived from each document indicates the likelihood that the document belongs to a particular topic, which is a numerical representation of what the review content is focused on. In particular, the BERTopic model has the advantage of better reflecting the contextual meaning of documents, allowing for more sophisticated analysis of textual data (Choi, 2024). The probability value of each topic derived from topic modeling indicates how central a feature or issue is in user reviews. Based on this, you can quantify the contribution of a topic to user satisfaction by analyzing whether it has a positive or negative impact on customer satisfaction (Choi, 2024).

Similarly, Park (2020) combined LDA topic modeling and Ordinary Least Squares (OLS) regression analysis on eco-friendly restaurant reviews to evaluate the impact of each topic on customer satisfaction, and Liu and Gao (2021) used LDA topic modeling and regression analysis on online psychological counseling reviews to evaluate the impact of each review's topic and emotional features on the usability of the review.

In addition, Yu et al. (2023) used LDA to thematize paper abstracts to analyze research trends in the field of artificial intelligence (AI), and used regression models to analyze the time series trend of each topic based on the derived topic distribution. The dependent variable was the

〈Table 4〉 Topics Probabilistic Regression Analysis study Comparison

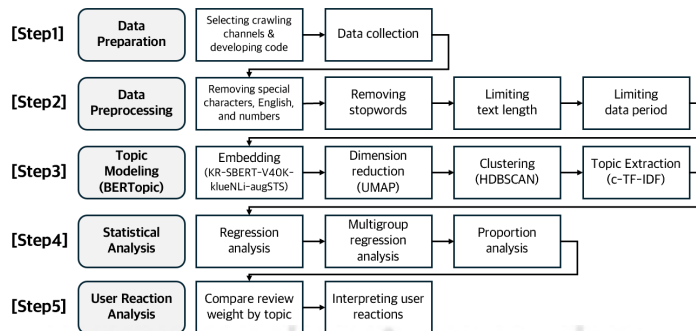
Researchers	Research subjects	Analytics resources	Analytics techniques
Eunhye Park	Green restaurant reviews	News Comments, Video subtitle	LDA, OLS Regression
Liu et al.	Online psychotherapy reviews	Counseling reviews text	LDA, Sentiment Analysis
Dejian Yu et al.	Research trends in AI	User Reviews	LDA, Linear Regression
This Research	Korea Financial Super App	User Reviews	BERTopic, Logistic Regression

change in the proportion of a particular topic from year to year. Linear regression analysis was used to evaluate the increase and decrease over time, making the regression model a quantitative tool for analyzing topic trends.

These previous studies analyzed factors affecting satisfaction and usability based on user reviews, but they only addressed the variation in functional satisfaction of a single app. In contrast, this study aims to empirically analyze the impact of the launch of a financial super app on user satisfaction between super apps and non-super apps by analyzing the topics that affect user satisfaction and comparing the proportion of reviews over time. In the process, this study utilized BERTopic to link topics and topic probability values reflecting the context between documents to the regression analysis to quantitatively check how user satisfaction changes in the super app environment even if the features are the same.

3. Methodology

This study analyzes user review data of S Financial Group's super app and its non-super apps (Pay, Trading, Insurance, and Banking) that provide more detailed features that are integrated in the super app. We used a methodology consisting of five steps: data preparation, data preprocessing, topic modeling, statistical analysis, and user response analysis to compare the difference in user reactions before and after the super-app launch and to investigate how the integrated features affected star ratings. The first step is to collect online reviews from the Google Play Store using Python. In the second step, we preprocess the collected data to perform topic modeling. In the third step, we apply the BERTopic model to extract the factors that contribute to star ratings in online reviews. In the fourth step, based on the extracted factors, OLS



[Figure 1] Methodology

regression analysis was performed using Python to check the relationship between the extracted topics and star ratings, and finally, the proportion of reviews by topic was categorized based on the changes before and after the super app launch to analyze user reactions.

3.1 Data Preparation

In this study, we collected user reviews posted on the Google Play Store, an Android-based application installation platform. The analyzed products include S Financial Group's super apps and non-super apps that provide integrated functions such as payment, securities, insurance, and banking in a single app. Reviews were collected until April 18, 2025 using the Python-based Google Play Scraper library. During this period, we obtained a total of 118,644 reviews. The details of the collected data are shown in <Table 5> below. The format of the collected reviews is shown in [Figure 2] and includes information such as star rating, date, review text, and number of review recommendations.

<Table 5> Data Collection Info

Types	Count
Super app	7,565
Non Super app	111,079
Total	118,644

★★★★★ 2024년 5월 15일

앱 리뷰에 별 1개 처음 달아봅니다. 통합앱이라고 광고도 많이 하고 추천에도 계속 떠 기존앱에서 바꿨는데 자주 사용하는 기능만 한 군데 모으려 했으나 중요한 기능은 없고 자잘한 기능만 모아 새로 만든 어플' 같은 느낌입니다. 페이지를 비롯하여 기존앱에서 사용하는게 더 편하거나 기존앱에는 있지만 통합버전에는 없는 기능도 더러있어 결국 기존앱을 다시 사용하게 되었습니다. 기존 UI에서 기능이 불편해지고 빠진거면 통합앱이라는 취지에 맞지 않는 거 아닐까요...? 처음에는 제가 사용을 잘 못해 그런건 줄 알았는데 다른 분들도 비슷한 댓글과 그 댓글에 공감 많이 많은걸로 봐선 어느 정도 어플이 개선되어야 하지 않나 하는 생각이 듭니다.

사용자 271명이 이 리뷰가 유용하다고 평가함

이 리뷰가 유용했나요? ☒ 예 ☐ 아니요

[Figure 2] Example of User Review

3.2 Data Preprocessing

In the data preprocessing phase, we transformed the data collected through Google Play Scraper into a format suitable for topic modeling. Since unstructured data such as text data needs to be transformed into an analyzable form for analysis (Li et al., 2021), we performed by removing stop words, punctuation, and numbers, and by converting all words to lowercase. This pre-processing step is important because the presence of such elements can affect the performance of text mining algorithms (Guo et al., 2017). Accordingly, the preprocessing was performed in five steps using several libraries in the Python program environment.

In the first step, we used regular expressions to remove foreign words, numbers, and special symbols, since we were analyzing online reviews written in Korean. In the second step, we removed words that could affect the results but were not meaningful, such as profanity, preposition, post-position etc. In the third step, we filtered out reviews with less than 10 characters to remove reviews that were irrelevant to the app or consisted of greetings, such as "thank you," "like it," or "best". Fourth, we included reviews collected through Google Play Scraper since 2012, but to clearly analyze the effect of super app conversion, we only included reviews written after January 1, 2023 to exclude the effects of changes in user behavior due to COVID-19 or the effect of app renewal. Fifth, we tokenized online reviews using the KIWI morphological analyzer for Korean. The existing BERTopic model uses sci-kit-learn's CountVectorizer internally, but it performs tokenization based on simple spacing, which has limitations for Korean analysis.

Therefore, this research utilized the KIWI analyzer, which has high disambiguation accuracy, typo and spacing error correction, and fast processing speed that reflects the context of Korean sentences, to perform more appropriate tokenization (Lee, 2024).

3.3 Topic Modeling

In this stage of the analysis, we used the BERTopic model to analyze the preprocessed review data. Compared to other methods, BERTopic demonstrates high topic coherence and competitive topic diversity (Grootendorst, 2022). The BERTopic model used in this research consists of model embedding, document clustering, and topic extraction. In the first step, a document converter was used to organize document-level embeddings. In this research, embeddings were performed using the pre-trained language model ‘KR-SBERT-V40K-klueNLI-augS TS’ for Korean text data. In the second step, UMAP was used to reduce the embedding dimensionality. In the third step, we used HDBSCAN as the clustering algorithm. We had planned to apply K-Means clustering, but the percentage of documents assigned -1, which is considered noise, was too high, so we replaced it. HDBSCAN is characterized by its ability to classify clusters of different densities, allowing it to effectively detect both small and large categories of noise (Lee et al., 2021). Finally, we used a variant of TF-IDF, Class-based TF-IDF, to extract the most important topics in the cluster.

When reducing the embedding dimension in UMAP, we set the following parameters: the number of nearest neighbors is 15, the number of dimensions is 5, the minimum distance between

clusters is 0.0, and the distance calculation method is ‘cosine’.

The initial results of HDBSCAN clustering showed a lot of overlap between similar topics and blurred topic distinctions. Therefore, we iteratively set the number of topics from 6 to 15 using the `reduce_topics()` function and reviewed the topic keywords and related review sentences. In addition, we calculated the coherence score for each number of topics, and the results are shown in <Table 6> below. The coherence score quantifies the degree of semantic interconnection among words within a given topic. It gauges the extent to which the words within a topic are meaningfully related, aligning with our human sense of a well-structured topic (Farea et al., 2024). However, a high coherence score does not necessarily provide the most appropriate topic structure for human interpretation. Hoyle et al. (2021) point out that automated metrics alone may be inadequate for model comparison and do not guarantee human interpretability. Therefore, we used the cohesion scores of the topics derived from each step, but also considered the overlap between topics, clarity of interpretation, and semantic distinction to select the final 10 topics.

<Table 6> Coherence score by number of Topics

Number of Topics	Coherence score
6	0.4508
7	0.4586
8	0.4429
9	0.4548
10	0.4565
11	0.4224
12	0.4260
13	0.4119
14	0.4083
15	0.4072

Coherence scores by number of topics can be found in <Table 6>.

3.4 Regression Analysis

Regression analysis is a statistical method used to define and quantify the relationship between variables, which can technically describe the association between the dependent and independent variables, and at the same time perform an adjustment function to control the influence of confounding variables or covariates. Due to these characteristics, this study adopts regression model as the basic analysis tool to analyze the impact of topic probability values derived from BERTopic on star ratings.

The independent variables for the analysis were derived from the BERTopic model. BERTopic is a model that extracts major topics from review texts and provides a probability value for each review to belong to a specific topic. In this study, the 10 clusters extracted by BERTopic were used as independent variables in the regression analysis. The dependent variable is the star rating of customer reviews on a scale of 1 to 5, and the OLS function provided by the Statsmodels library in Python was used for the analysis. To ensure the validity of the regression model, we first checked for multicollinearity among the topics using VIF (Variance Inflation Factor). It considered that if the independent variables included in the regression model are highly correlated with each other, the regression coefficient estimates may be distorted or the results may be unreliable (Harrell, 2001; Hosmer et al., 2013).

We constructed three regression models. The first model (1) is designed to analyze how the probability of each review belonging to a specific

topic affects its score. The dependent variable is the star rating of a user review, and the independent variables are the probability values of the 10 topics extracted through BERTopic. The regression coefficient indicates the relative influence of each topic on the star rating. The purpose of this model is to understand the influence of the topics themselves based on the total reviews without separating super apps from non-super apps.

$$score_i = \sum_{k=0}^9 \beta_k \cdot topic_{ki} + \epsilon_i$$

In the second model (2), we consider the main effect term of superapp status on star ratings. is a dummy variable indicating whether the review is from a superapp, where 1 means superapp and 0 means non-superapp. The regression coefficient γ indicates the independent effect of superapp status on star ratings, which may reflect that the level of user ratings, i.e., star ratings, varies depending on whether a superapp is available or not.

$$score_i = \sum_{k=0}^9 \beta_k \cdot topic_{ki} + \gamma \cdot superAppYN_i + \epsilon_i$$

The third model (3) includes an interaction term to analyze how the influence of a topic varies depending on whether it is a superapp or not. is the coefficient of the interaction term between each topic's probability value and its superapp status, reflecting the fact that the same topic may have different effects on star ratings in superapps and non-superapps. This allows us to see how the app structural characteristic of being a super-app moderates the interpretation of user satisfaction for a particular topic.

$$score_i = \sum_{k=0}^9 \beta_k \cdot topic_{ki} + \gamma \cdot superAppYN_i \\ + \sum_{k=0}^9 \delta_{ki} \cdot (topic_{ki} \cdot superAppYN_i) + \epsilon_i$$

4. Analysis and Results

4.1 Frequency Analysis Results

In this research, we reviewed the results of a basic analysis of the frequency of occurrence of words in online reviews to identify overall keywords and trends. The frequency analysis of the words extracted from super apps and non-super apps showed that super apps were dominated by words such as ‘use’, ‘function’, ‘install’, ‘authentication’, and ‘integration’ while non-super apps were dominated by words such as ‘use’, ‘authentication’, ‘sign up’, ‘event’, and ‘pay’, with the top five most common words being ‘user’, ‘use’, ‘customer’, ‘function’, and ‘service’. This analysis shows that words like ‘use’, ‘features’, and ‘service’ are common among both super apps and non-super apps, indicating that usability and functionality-related topics are dominant in online reviews. These results suggest that usability and functional ease of use are key concerns for users of both super and non-super apps.

4.2 Topic Analysis Results

BERTopic models can iteratively analyze data to identify topics in topic modeling (Jeong et al., 2023). In performing BERTopic based topic modeling, we predefined the number of topics to be 10 in consideration of the analysis purpose and domain characteristics, rather than using HDBSCAN’s density-based clustering method to automatically derive the number of topics.

According to Bahja and Safdar (2020), effective topic classification can be confirmed if the inter-topic distance map shows that topics are well separated and independent from each other. Through this, we derived 10 topics that contribute to application star ratings in this research. [Figure 3] is an IDM visualization of the keywords of the topics analyzed by the collected reviews and the number of reviews assigned to each topic, and the feature names that define the derived words are shown in <Table 7>. To define each topic, we referenced the original text of the review assigned to that topic.

Topic_0 is defined as ‘Identity Verification Service’ based on the original review text. An identity verification service is a service that verifies whether a user is the real person using electronic information in a real-time processing system when an online user uses Internet services such as registering on an Internet site, changing a password, or writing on a bulletin board (Shin et al., 2015). It is an essential service to verify the identity of the user in order to use financial-related services.

Topic_1 is defined as ‘Stock Trading Function’ based on the original review text. According to the Korea Securities Depository (2025), the number of individual investors in Korea is 14.1million, which is about 33% of the Korean population, so there is a high level of social interest in stock investment. The topic is related to the main transaction functions within the app, such as buying and selling stocks, searching for stocks, and checking balances, and refers to the core and frequently used functional areas from the user’s perspective.

Topic_2 is defined as “Simple Payment Service” based on the original review text. When using

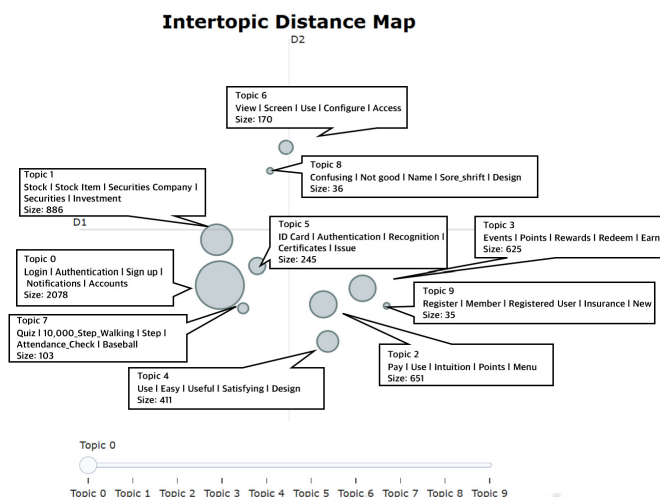
an online payment service, it refers to a feature that allows users to make simple and secure payments without entering the card number directly

by utilizing a bank or card app, rather than entering the card number directly.

Topic_3 is defined as 'Event' based on the origi-

<Table 7> UI/UX Elements Defined based on Representative Keywords and Reviews Per Topic

Topic	Elements	Keyword	Original review content
0	Identity Verification Service	Login, Authentication, Sign up, Notifications, Accounts	I reinstalled the app and the problem persists. I wish there was an option like text verification or simple verification.
1	Stock Trading Function	Stock, Stock Item, Securities Company, Securities, Investment	Reacting to short-term trades has become significantly more difficult
2	Simple Payment System	Pay, Use, Intuition, Points, Menu	I mainly use my S Bank card and S Pay is easy to see and use.
3	Event	Events, Points, Rewards, Redeem, Earn	It's not hard to use for people our age, and the rewards and events are fun and challenging.
4	Menu UI	Use, Easy, Useful, Satisfying, Design	More content, more offers, and a simpler menu design What a great makeover.
5	ID Card Recognition Function	ID Card, Authentication, Recognition, Certificates, Issue	We need to get the real name verification straight, or we need to get rid of the ID verification crap.
6	Home Screen UI	View, Screen, Use, Configure, Access	I think I'm more comfortable using it because I can see my favorite menus at a glance on the first screen.
7	Daily Participatory Interaction	Quiz, 10,000_Step_Walking, Step, Attendance_check, Accounts	You get points for using the card, you get points for writing reviews, you get points for attendance, you get points for taking quizzes, and you get points for app.
8	App Icon Design	Confusing, Not good, Name, Sore_shrift, Design	S app names are confusing because they look alike
9	Sign-up Process Function	Register, Member, Registered User, Insurance, New	I signed up, but every time I run it, it keeps asking me to sign up again



[Figure 3] IDM Result

nal review text. It refers to various types of promotional activities such as giveaways and fee reductions to attract users and encourage them to continue using the service.

Topic_4 is defined as ‘Menu UI’ based on the original review text. This refers to a menu-based user interface that allows users to access and navigate various functions within the app, and is a factor that closely affects the flow of app usage and overall satisfaction.

Topic_5 is defined as “ID Card Recognition Function” based on the original review text. This is a key technology used in the initial authentication stage for non-face-to-face account opening, which enables the app to accurately recognize the user’s ID card image and process it without errors.

Topic_6 is defined as ‘Home Screen UI’ based on the original review text. It is the first screen that users encounter when they first launch the app, and it is a factor that determines the first impression of the main service.

Topic_7 is defined as ‘Daily Participatory Interaction’ based on the original review text. Park et al.(2025) found that SNS are making it increasingly difficult for users to leave the service through various interaction design elements within the service. The S Financial group apps also include elements that induce repetitive engagement, such as attendance check, baseball, and pedometer, and this feature is likely to be used as a strategic means to induce users’ habitual use of the app and increase the rate of app revisits.

Topic_8 is defined as “App Icon Design” based on the original review text. This is the first visual element a user encounters when installing or launching an app, and refers to a symbolic image

that visually expresses the identity of the app, such as color, shape, and design.

Topic_9 is defined as ‘Sign-up Process Function’ based on the original review text. This is a basic and essential procedure for using the service, including the initial registration process such as entering user information, verifying identity, and agreeing to terms and conditions.

4.3 Regression Analysis Results

The results of the multicollinearity analysis among the functional factors proposed in this research are shown in <Table 8>. When evaluating multicollinearity issues, it is appropriate to look for a VIF (Variance Inflation Factor) of 3.3 or 5.0 or less (Kock, 2015), and since the values of VIF are below 3.3 for all variables, we confirmed that there is no multicollinearity among the variables.

<Table 8> Multicollinearity by Topic

Topic	VIF	1/VIF
Topic_0	2.0477	0.4883
Topic_1	1.5589	0.6415
Topic_2	1.5295	0.6538
Topic_3	1.4245	0.7020
Topic_4	1.2779	0.7825
Topic_5	1.1985	0.8344
Topic_6	1.0975	0.9111
Topic_7	1.0416	0.9601
Topic_8	1.0275	0.9733
Topic_9	1.0175	0.9828

The first regression model has an explanatory power (R^2) of 0.706, indicating that the topic probability values derived from BERTopic explain about 70.6% of the user ratings. The F-statistic

for the overall regression is 2032 ($p < 0.001$), indicating that the model is statistically significant. The regression coefficients for each topic are all significant at the $p < 0.001$ level, indicating that a particular topic has a significant impact on user ratings.

The second model extends the analysis by adding superapp status, and the explanatory power (R^2) is 0.707, a slight increase from the first model. The F-statistic is statistically significant at 1848 ($p < 0.001$), and the superAppYN variable, which indicates whether a superapp is a superapp, has a regression coefficient of 0.1033 ($p = 0.048$), indicating that superapp status has a significant positive effect on user ratings.

In the third model, we included an interaction term between topic probability and superapp status, which slightly increased the R^2 to 0.721. The overall regression model was significant with an F-statistic of 1040 ($p < 0.001$), suggesting that the addition of the interaction term contributed to improved explanatory power. The interaction coefficients were mostly significant in the negative direction, and the p-values for each item were mostly less than 0.001, with the exception of “Topic_9 $\times$ SuperApp”, which showed a slightly lower level of significance at $p = 0.040$. This shows that there is a moderating effect where the same topic has a significantly different impact on the star rating depending on whether it is a superapp or not.

Next, the regression results, shown in <Table 9>, are significant in all models. All topics have a positive effect in the basic model (1), but the regression results in <Table 9>, which considers the main effect term of super apps, are significant in all models. All topics show a positive effect in the basic model (1), but in the model (2), which

considers the main effect of super app, satisfaction shows a slight decrease, and in the interaction model (3), which reflects the super app environment, the interaction coefficient is negative for all topics. This shows that the same UI/UX elements cause lower star ratings when they work in a superapp. This result can be interpreted as not a problem with the elements themselves, but rather that the superapp structure leads users to form higher expectations. In this regard, Nurqamarani et al. (2020) empirically demonstrated that superapp users expect higher quality and simplicity, and that satisfaction can drop sharply when these expectations are not met, and regression analysis shows that this is the case.

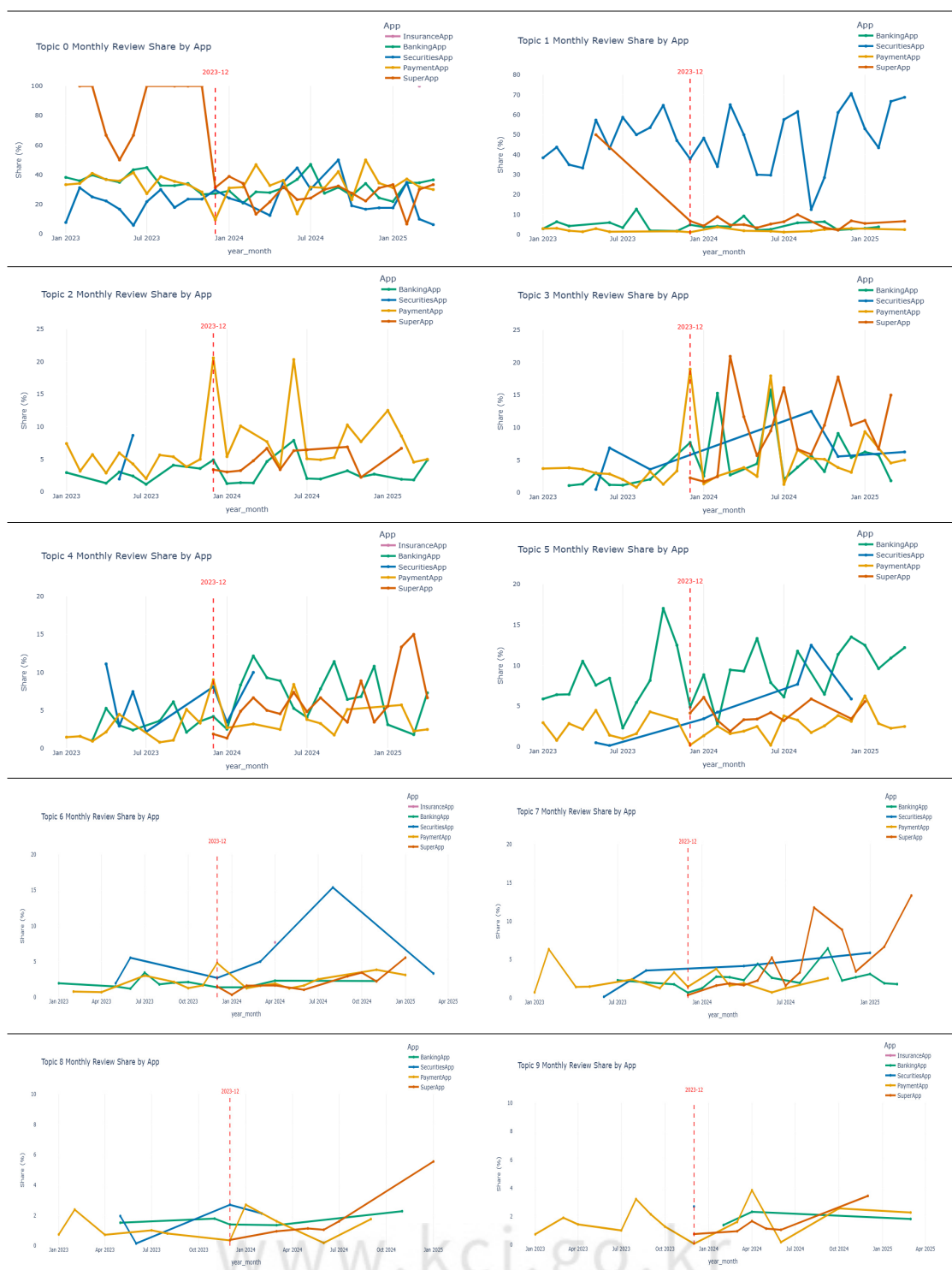
The detailed analysis by topic can be interpreted in conjunction with the analysis of the change in the proportion of reviews by topic in <Table 10>. Considering the movement of topics over time before and after the introduction of super apps, we were able to categorize topics into three groups. The first group (Group_1) is Topic_1, Topic_2, Topic_5, and Topic_6, where we can observe that the importance of existing features remains significant in the non-super app environment even after the introduction of super apps. The second group (Group_2) is Topic_3 and Topic_7, which showed that the influence of topics became more prominent in super apps after the emergence of super apps, and consisted of topics that aimed to provide new value to users based on the integration and scalability of super apps. The last group (Group_3) is composed of Topic_0, Topic_4, Topic_8, and Topic_9, which can be seen to influence user satisfaction in the entire app environment regardless of whether a super app has appeared.

〈Table 9〉 Regression Results

Topic	(1)	(2)	(3)
Topic_0	1.9428*** (0.0404)	1.9179*** (0.0422)	1.9940*** (0.0438)
Topic_1	2.8429*** (0.0642)	2.8351*** (0.0642)	2.8894*** (0.065)
Topic_2	4.7448*** (0.0707)	4.7347*** (0.0708)	4.9009*** (0.0718)
Topic_3	5.3267*** (0.0806)	5.3088*** (0.0811)	5.3537*** (0.085)
Topic_4	5.4859*** (0.1054)	5.4697*** (0.1056)	5.4526*** (0.1099)
Topic_5	1.3633*** (0.1191)	1.3373*** (0.1197)	1.3602*** (0.1302)
Topic_6	4.3842*** (0.1690)	4.3763*** (0.169)	4.5053*** (0.1713)
Topic_7	3.4118*** (0.2795)	3.3668*** (0.2803)	3.5660*** (0.3458)
Topic_8	2.5885*** (0.3216)	2.5711*** (0.3216)	2.6008*** (0.3369)
Topic_9	2.3812*** (0.4143)	2.3572*** (0.4144)	2.2553*** (0.4575)
SuperApp		0.1033* (0.0522)	2.4484*** (0.1286)
Topic_0_x_SuperApp			-3.1153*** (0.1829)
Topic_1_x_SuperApp			-3.1707*** (0.2742)
Topic_2_x_SuperApp			-5.1889*** (0.3065)
Topic_3_x_SuperApp			-3.0012*** (0.2689)
Topic_4_x_SuperApp			-2.6468*** (0.346)
Topic_5_x_SuperApp			-2.7512*** (0.3236)
Topic_6_x_SuperApp			-4.2243*** (0.6393)
Topic_7_x_SuperApp			-3.0965*** (0.5795)
Topic_8_x_SuperApp			-2.8682** (0.9339)
Topic_9_x_SuperApp			-2.0222* (0.9835)

*(): S.E., *p<0.05, **p<0.01, ***p<0.001.

<Table 10> Visualization of review weight by topic



<Table 11> Results of a Regression Analysis Performed with and Without Superapps

Topic	SuperApp	Non SuperApp
Topic_0	1.7918*** (0.083)	1.9940*** (0.0455)
Topic_1	2.2633*** (0.2124)	2.8894*** (0.0675)
Topic_2	2.8852*** (0.2278)	4.9009*** (0.0746)
Topic_3	5.2059*** (0.1903)	5.3537*** (0.0883)
Topic_4	5.7592*** (0.2666)	5.4526*** (0.1142)
Topic_5	1.4354*** (0.2364)	1.3602*** (0.1353)
Topic_6	2.8897*** (0.5538)	4.5053*** (0.178)
Topic_7	3.1966*** (0.4079)	3.5660*** (0.3592)
Topic_8	2.6224*** (0.791)	2.6008*** (0.3499)
Topic_9	2.8278*** (0.793)	2.2553*** (0.4753)

(): S.E., * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Based on this categorization of reviews by topic, the interpretation of each topic is as follows.

For Group_1's Topic_1 (Stock Trading Function), the regression analysis shows that the coefficient (β_1) in the base model is 2.8429, and the coefficient (β_2) in the model with the SuperApp variable is 2.8351, which maintains a similar level of moderate satisfaction. However, the interaction term between Topic_1 and SuperApps is -3.1707, with a negative direction, suggesting that satisfaction with the feature is significantly reduced in the SuperApp environment. The regression results in <Table 11>, which completely separates super apps and non-super apps, also show that satisfaction is lower in the super app environment, with an 2.2633 for super apps and 2.8894 for non-super apps. The topic also shows that despite

the release of super apps with stock app features during the analysis period, reviews related to the stock trading feature still remain overwhelmingly in stock apps, suggesting that the stock trading feature provides a more positive user experience in existing stock apps than in super apps. In other words, Topic_1 is a feature that remains strong in stock apps (non-super apps) even after the launch of the super app, which can be interpreted as a result of the fact that stock apps have been providing stock trading services that reflect user feedback for a long time. Chang et al. (2019) argue that the mobility and ubiquity of MTSs are highly compatible with the characteristics of the stock market, which requires real-time stock trading and information provision, enabling their continued growth. From this perspective, stock apps have advanced their functions as MTSs based on accumulated user experience and have tried to provide satisfactory services to users. On the other hand, the Super App is currently in the initial stage of integrating only some stock trading functions, which requires continuous monitoring and strategic judgment on what auxiliary functions to add and subtract to provide users with a more satisfying experience.

Next, when we look at Topic_2 (Simple Payment System), the coefficient (β_1) in the basic model is 4.7448, and the coefficient (β_2) in the model with the super app variable is 4.7347, which is slightly lower. However, the interaction term between Topic_2 and SuperApps is -5.1889, which is the largest negative value among all topics. As shown in <Table 11>, the coefficient of 4.9009 was very high for non-super apps, but only 2.8852 for super apps, indicating a clear difference in satisfaction between the two groups. This trend is also evident in the change in review

weight. While there were a few mentions of simple payment services in the super app experience, the overall weight of reviews was very much in favor of PayApp. This suggests that when PayApp was provided as a simple payment app, users were satisfied with its clear accessibility and high concentration of features, but when SuperApp appeared and payment functions were expanded to SuperApp as well as PayApp, user satisfaction decreased due to the complexity of the access path and the decrease in feature awareness. In addition, the dilemma between convenience and complementarity in the information system or information technology use environment is also present in the mobile payment service environment (Kim et al., 2017), which suggests that the characteristics of super apps that provide multiple services at once may lead to a paradoxical situation where users expect high convenience, but the complexity of the menu structure and enhanced security procedures may lead to a decrease in user experience.

Next, we looked at the regression results of Topic_5 (ID Card Recognition Function), and found that the coefficient (β_1) of the basic model was 1.3633, while the coefficient (β_2) of the model with the Super App variable was 1.3373, showing the lowest satisfaction among all topics. In addition, the interaction term between Topic_5 and Super App was -2.7512, showing that the satisfaction with the feature in Super App is in a negative direction. <Table 11> shows that the regression coefficient is 1.4354 for super apps and 1.3602 for non-super apps. In the analysis of the proportion of reviews, banking and securities apps have a relatively high proportion of reviews. This can be interpreted as a phenomenon caused by the difference in functional structure between su-

per apps and non-super apps. Guan et al. (2024) state that effective fraud detection and analysis of government-issued identity documents—such as passports, driver's licenses, and ID cards—are essential for preventing identity theft and enhancing the security of online platforms. As such, the ID card recognition function is an essential step in accessing financial services and can be seen as a topical element that most users will encounter, regardless of whether they are in a superapp or non-superapp. However, differences can be seen in the placement of the feature and the user journey. In the case of super apps, users who have already opened an account often access them for a variety of other services, rather than to use the services that require identity verification, and even if they want to open a new account, they have to go through a multi-click process of 'Menu', 'Finance', and 'Open Account'. On the other hand, in the case of banking or securities apps, the main purpose of the app is financial transactions, so the ID card recognition function is placed at the beginning of the app usage as part of the core flow such as account opening. Due to these structural differences, even if the same functionality is provided, the accessibility, frequency of exposure, and awareness of the feature from the user's perspective will vary across platforms, and the process of reaching the feature in a superapp may be perceived as complex, leading to low satisfaction. As a result, ID card recognition function is a feature whose evaluation depends on the frequency of exposure and procedural intuitiveness along the user journey, apart from technical perfection, and in this regard, it is interpreted that it remains a major topic of user interest in banking and securities apps.

Finally, looking at the regression results for

Topic_6 (Home Screen UI) in Group_1, the coefficient (β_1) of the base model is 4.3842, while the coefficient (β_2) of the model with the super app variable is 4.3763, which is negative. In addition, the interaction term between Topic_6 and Super Apps is -4.2243, indicating that satisfaction with the feature in the Super Apps environment is significantly negative. <Table 11> shows that the regression coefficient is 2.8897 for super apps and 4.5053 for non-super apps, showing a stark difference in satisfaction, just like Topic_2. When analyzing the proportion of reviews, the topic was mainly concentrated on the stock app, which can be interpreted as the app intuitively lays out various stock-related information such as buying and selling stocks, managing holdings, and checking market prices. On the other hand, the super app offers limited stock information features, and the home screen is designed to provide various services rather than focusing on stock information. For users who mainly use the stock function, it is believed that the inconvenience of not being able to intuitively access the desired information directly from the home screen of the super app leads to a decrease in satisfaction. In other words, compared to the securities app, which has a function-focused UI that provides only securities-related information on the home screen, the variety of information provided by the super app seems to have a negative impact on the accessibility and efficiency of information for securities app users.

Now, when we look at Group_2, we can see that these topics are based on the structural strength of SuperApps, which is the ability to easily expose and connect various functions in one app. First, the regression results for Topic_3 (Event) show that the coefficient (β_1) in the basic

model is 5.3267, while the coefficient (β_2) in the model with the super app variable is 5.3088. On the other hand, the interaction term between Topic_3 and Super Apps is -3.0012, indicating that satisfaction with the feature tends to decrease slightly in the Super Apps environment. In <Table 11>, the regression coefficient was 5.2059 for super apps and 5.3537 for non-super apps. When analyzing the proportion of reviews, we found a significant increase in super apps. This suggests that the event factor is recognized as a factor that increases satisfaction, but it does not provide as much differential utility as expected in the super app environment. The super app user does not have to search and download multiple applications for different tasks, as all the services are available in a single application, thereby saving time and effort (Amit et al., 2023). This structure of a superapp can be viewed positively in terms of improving accessibility and increasing consumer satisfaction by consolidating a large number of disparate events. However, the actual analysis shows that the event element in superapp environments does not significantly provide users with clear convenience or differentiated satisfaction compared to non-superapp environments. In other words, the information integration itself may be a positive factor, but it does not fully fulfill user expectations such as differentiation.

Next, the regression results for Topic_7 (Daily Participatory Interaction) show that the coefficient (β_1) in the base model is 3.4118, while the coefficient (β_2) in the model with the superapp variable is 3.3668, indicating a moderate level of satisfaction. On the other hand, the interaction term between Topic_7 and Super Apps was -3.0955, indicating that satisfaction with the fea-

ture tended to decrease significantly in the Super Apps environment. In <Table 11>, which analyzes super apps and non-super apps separately, we can see that the regression coefficient is 3.1966 for super apps and 3.5660 for non-super apps. These numbers indicate that the feature is not causing clear dissatisfaction, but that the perceived satisfaction is relatively low compared to expectations. The analysis of the proportion of reviews shows that the most significant increase is after the launch of the super app, which indicates that various gamification elements such as baseball, quizzes, all views, and attendance checks are more actively introduced in the super app environment. By integrating game mechanics naturally into the user experience, marketers can create more enjoyable interactions that keep customers coming back for more, according to Chen (2023). In fact, we observe that S Financial Group's super app strategically increases time in app and retention by introducing various gamification-based features such as balance games, quizzes, baseball, etc. to motivate repeat usage and build daily habits. However, based on the regression analysis, it seems that while these factors may have contributed to increasing user acquisition and dwell time as a marketing strategy, there is a limit to the extent to which they translate into qualitative user experience satisfaction.

Finally, we can see that the topics in Group_3 have a universal impact on user satisfaction regardless of the specific app type and consist of the basic features of the app. First, the regression analysis of Topic_0 (identity verification service) shows that the coefficient of the basic model (β_1) is 1.9428, and the coefficient of the model including the superapp variable (β_2) is 1.9179, which shows a low level of satisfaction. In addition, the inter-

action term between Topic_0 and superapp is -3.1153, which shows that satisfaction with the feature tends to decrease further in the superapp environment. In <Table 11>, the regression coefficient of 1.7918 for super apps and 1.9940 for non-super apps showed low satisfaction in both environments, but the dissatisfaction was more pronounced in super apps. This shows that identity verification is a problematic user experience across apps, but it's even worse in super apps. In recent years, identity verification services have been introduced in connection with new devices and services, including new financial transactions such as fintech and simple payment services (Shin et al., 2017). However, in this research, it is characterized by a greater negative perception of the authentication function in the superapp environment. Yang et al. (2023) stated that in the superapp ecosystem, it is very important to balance security and usability to optimize user experience while protecting user data and privacy. However, due to the integrated nature of super apps that provide multiple functions, the security level is bound to be relatively high, which complicates the authentication process and increases the likelihood that users will experience greater inconvenience. As a result, even for the same authentication function, the complex service flow and high security requirements in super apps lead to a decrease in usability, which can be interpreted as a lower satisfaction rate compared to non-super apps.

Next, the regression analysis of Topic_4 (Menu UI) showed that the coefficient of the basic model (β_1) was 5.4859, and the coefficient of the super app variable (β_2) was 5.4697, which was the highest satisfaction among all topics. Also, the interaction term between Topic_4 and super app was -2.6468, which was relatively lower than

other topics. In <Table 11>, the regression coefficient of 5.7592 for super apps and 5.4526 for non-super apps shows the highest satisfaction in both environments. Murano and Sander (2016) found that the strongest contributor to improved performance was a menu that was as simple and well-organized as possible. This can be interpreted as a result of superapps overcoming structural limitations that can lead to complexity as they integrate various features, and striving for user convenience when organizing menus.

Next, the regression analysis of Topic_8 (App Icon Design) showed that the coefficient (β_1) of the basic model was 2.5885, and the coefficient (β_2) of the model including the super app variable was 2.5771, showing low satisfaction. In addition, the interaction term between Topic_8 and super apps was -2.8682, showing a relatively low decrease. In <Table 11>, the regression coefficient was 2.6224 for super apps and 2.6008 for non-super apps, and it is characterized that non-super apps showed lower satisfaction. According to Goud (2022), visual branding elements such as logos not only attract attention but also play a crucial role in helping consumers recognize and associate with a particular brand. However, since the launch of the super app, the visual identity of S Financial Group's apps has become similar and the boundaries between affiliated apps have become blurred, which has led to user confusion, and it can be interpreted that users are not satisfied with the experience.

Finally, the regression analysis of Topic_9 (Sign-up Process Function) showed that the coefficient (β_1) of the basic model was 2.3812, and the coefficient (β_2) of the model with the super app variable was 2.3572, which showed a decrease in satisfaction. In addition, the interaction term

between Topic_9 and super app was -2.0222, which showed the lowest decrease in satisfaction. <Table 11> shows that the regression coefficient was 2.8278 for super apps and 2.2553 for non-super apps, indicating that non-super apps showed lower satisfaction. In other words, although the absolute satisfaction level is relatively better in super apps, the interaction term is in a negative direction, suggesting that the super app structure itself is an environmental factor that can further reduce the evaluation of the membership function. Wang (2018) suggests that realizing the "sign up" and "sign in" functionality has a significant business impact and requires careful and deliberate design because it can have a significant impact on so many business functions. From this perspective, signup functionality is not just a super-app issue, but should be designed with user experience in mind for other non-superapps as well. However, it can be interpreted that the mismatch between the expectation that users will be able to use all services in the superapp environment and the reality that there are features in the non-superapp that are not available and can only be accessed by signing up for a separate subscription led to a negative user experience.

5. Conclusion

5.1 Academic Implications

Existing user experience (UX) research has mainly focused on qualitative approaches or single-point-in-time evaluations, limiting the ability to empirically analyze temporal changes in user responses or contextual differences across features. In particular, the discussion of how the multi-dimensional attributes that comprise UX undergo

structural changes during service transitions has been relatively neglected in the literature. By applying BERTopic, a state-of-the-art natural language processing technique that reflects context based on online review data, this study attempts to extend the methodology to quantitatively analyze user responses. In particular, by empirically quantifying user responses by topic and analyzing the changes in perceptions of key UI/UX elements due to structural changes in the financial app ecosystem before and after the launch of super apps, this study empirically reveals the UI/UX impact of super apps on non-super apps, which has been overlooked in previous studies. This study makes a significant academic contribution in that it provides an analytical framework that can be used to set strategic priorities by function in future superapp-based UI/UX design and set structural improvement directions. Furthermore, this study makes an important academic contribution in that it quantitatively demonstrates the structural changes and impacts of superapp adoption through a comparative analysis based on a time breakpoint that existing studies have not attempted.

5.2 Practical Implications

From a practical perspective, this study identifies UI/UX elements that users perceive as important in the financial app environment and quantitatively shows the impact of UI/UX elements on user satisfaction in the app ecosystem changed by the launch of super apps, providing practical insights for app planning and operation strategies. The analysis shows that in the superapp environment, expectations of integrated services are generally higher, and user satisfaction ratings tend to decrease. This highlights

the importance of managing expectations and applying consistent UX design principles, as more features do not necessarily translate to user satisfaction.

Furthermore, by analyzing how certain elements are considered a higher priority in non-superapp environments and user satisfaction is more positive in those environments, it suggests the need for a different approach when updating superapps and non-superapps in the future, and can be used as a reference point for practical strategy in the planning and operation of financial superapps. For example, in the non-superapp environment, users were relatively satisfied with factors such as “ease of login process,” “speed of account transfer,” and “clarity of information organization on the screen,” while in the superapp environment, users complained about the lack of consistent UX design compared to the diversity of functions. As a result, practitioners can set strategic improvement directions for super apps, such as securing UI consistency across functions, designing information centered on main functions, and minimizing frequent authentication requirements.

As such, this study provides a reference point for setting priorities for UI/UX improvement in practice by comparing the relative influence of each topic through empirical analysis of user satisfaction by major topics. In addition, the satisfaction coefficients derived from this study provide a numerical representation of the relative impact of satisfaction by feature, which can be used by app planners and UI/UX designers to make data-driven decisions about which features should be prioritized for redesign. This has a practical appeal in that it enables strategic, user-centered improvements that go beyond aesthetic design or one-off feature additions.

5.3 Limitations and Further Research

While this research may have academic and practical implications, it has the following limitations.

First, this research collected and analyzed online reviews written on the Google Play Store. It does not reflect the opinions of iOS users using iPhones. As of Q4 2024, Apple's share of the Korean smartphone market is 39% and rising, while Samsung's share remains at 60%. This shows that the percentage of iPhone users in South Korea is by no means low. With this in mind, we tried to collect app store review data, but we were unable to retrieve the full review data, so we excluded reviews written on the app store from our research. However, the proportion of iPhone users in Korea is increasing, and banking app usage encompasses all age groups and various user types, so an analysis biased toward a specific operating system may not fully reflect the needs and satisfaction of users across the board. It's important to include data from a variety of platforms, especially since user experience and usage can vary by operating system. If it is possible to collect reviews written in the app store, it should be included in the research.

Second, this research collected and analyzed online reviews. Online reviews may be biased in terms of sample representativeness because they only reflect the opinions of users who voluntarily write reviews. In particular, real users who leave reviews are most likely to have had a very positive or negative experience, which can lead to self-selection bias in review data (Bhole & Hanna, 2017). To compensate for these limitations, future research should include qualitative research methods, such as surveys or interviews,

that can reflect a variety of user opinions. In addition, temporal bias may also affect the interpretation of this research. The collected reviews were originally included from 2012, but in order to more clearly analyze the super app conversion effect, this research only included reviews written after January 1, 2023 to control for external factors such as rapid changes in user behavior due to COVID-19 and the renewal and rebuilding of the apps themselves. While this is beneficial for clarifying the focus of the analysis, it is a limitation of time-series analysis in that it only reflects the user experience after a certain point in time, which does not allow for long-term changes in user responses or comparisons before and after the introduction of super apps. Therefore, in future research, it is necessary to analyze the change trend of each point in time from a long-term perspective, or to design a design that can compare the user experience before and after the introduction of the super app.

Third, this research focuses on the super app of S Financial Group, one of Korea's leading financial groups, and the non-super app that provides functions that are integrated in the super app. In addition to S Financial Group, there are various other large financial holding companies such as K Financial Group, H Financial Group, and N Financial Group, which are also operating super apps as part of their digital transformation strategies. The implementation method and customer contact strategy of each financial group's super app may vary from group to group, and user perception and satisfaction may vary depending on the technical structure, customer management system, and brand image of the integrated approach. In addition, the reviews used in this research are limited to the service experience of

S Financial Group, which does not fully reflect how customers of various financial groups perceive super apps. Therefore, a broader and more generalizable insight can be drawn by analyzing the super apps of various financial companies.

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